KALYAN KUMAR DOPPALAPUDI

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EDUCATION

State University of New York at Buffalo Aug 2024 – Dec 2025
Master of Science in Computer Science
GPA: 3.76/4.0
Relevant Courses: Machine Learning, Deep Learning, Data Intensive Computing, Computer Security, Operating Systems

SRM University AP(India) Aug 2020 – June 2024
Bachelor of Technology in Computer Science
Relevant Courses: Data Structures, Design and Analysis of Algorithms, Object-Oriented Programming (Python, C++), Database Management Systems, Operating Systems, Computer Networks.
CGPA: 8.37/10

TECHNICAL SKILLS

Programming languages: Python, C, C++, Java, JavaScript
Databases & Data Tools: PostgreSQL, MySQL, MongoDB
Data & Analytics: Pandas, NumPy, Power BI, Tableau
Backend & Web: Node.js, Express.js, REST APIs, Microservices
Cloud & Tools: AWS, Docker, Kubernetes, CI/CD Pipelines
Other: Linux, Git, Agile/Scrum, Unit Testing

EXPERIENCES

Software Engineer | JUBLIEE INFORMATION TECHNOLOGY PVT LTD Jan 2023 – June 2024

- Developed scalable backend services and RESTful APIs using Node.js and Python, implementing core business logic and integrating with PostgreSQL, MySQL, and MongoDB to support high-volume, fault-tolerant applications with modular architecture.
- Optimized database operations and queries, including indexing and schema improvements, enhancing query efficiency, data consistency, and system performance by 30–40%, while ensuring reliability for concurrent workloads.
- Debugged, refactored, and deployed production-ready features in Dockerized environments, implemented logging, monitoring, and automated tests, contributing to resilient, maintainable, and stable releases across development, staging, and production environments.

Salesforce Developer Intern | Salesforce July 2022 – Sep 2022

- Built custom automation workflows using Apex classes, triggers, and Lightning components to streamline business processes and ensure reliable functionality.
- Designed and optimized SOQL/SOSL queries to efficiently process large datasets, improving query performance and maintaining system data integrity.
- Collaborated remotely with team members to gather requirements, troubleshoot code issues, and deliver stable features within deadlines.

PROJECTS

E-Learning Web Application

- Developed a full-stack web application using HTML, JavaScript, PHP, and Node.js, enabling structured user and module management.
- Implemented core functionalities including assignment submission, evaluations, quizzes, and content management, with MongoDB for secure data storage.
- Performed unit testing using Chai to validate all web functionalities, improving reliability and overall system organization prior to deployment.

Customer Churn Prediction Using Advanced ML Models

- Built an end-to-end churn prediction system using Python, Pandas, and NumPy for preprocessing, feature engineering, and model evaluation on large datasets.
- Developed advanced models including Random Leaf Model (RLM), Logit Leaf Model (LLM), and neural network ensembles, exposed via REST APIs built with Node.js and Express.js.
- Optimized data storage and retrieval using PostgreSQL, MySQL, and MongoDB (indexing, normalization, SQL tuning) and containerized the application with Docker for scalable deployment.

Real-Time Object Detection and Tracking

- Built a real-time object detection and tracking system using Python and OpenCV, enabling detection, localization, and tracking of multiple objects from video streams.
- Designed modular and reusable detection logic on Linux to maintain consistent object identification across varying lighting, camera angles, and environmental conditions.
- Implemented modular and reusable detection algorithms on different systems, ensuring robust and accurate object identification across varying lighting conditions, camera angles, and dynamic environmental changes, while enabling easy integration and scalability for future enhancements.

Efficient Task Scheduling for Latency-Sensitive Tasks in Edge Computing

- Analyzed latency, throughput, and resource utilization challenges in edge-cloud computing environments, benchmarking state-of-the-art task scheduling algorithms under real-time workloads.
- Designed and implemented optimization strategies for task allocation and resource management, improving system throughput and reducing end-to-end latency by 15–20%.
- Developed simulation pipelines to validate algorithm performance, ensuring scalability, efficiency, and reliability of distributed, latency-sensitive systems in practical scenarios.

ACHIEVEMENTS

- Filed patent: “Random Leaf Model (RLM)” a hybrid machine learning model for real-time predictive analytics, enhancing accuracy and efficiency (In process).
- Earned HackerRank certifications in Problem Solving (Intermediate/Advanced), SQL, Python, and Java.